Type MS · Expanding a bracket with a negative or fractional power

Answer 1 of 18 · Question 9 part a · 3 marks

2020 JAN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Number		
9 (a)	$1+\frac{1}{2}(-4x)+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)(-4x)^2}{2!}+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(-4x)^3}{3!}$	M1
	$1-2x-2x^2-4x^3$	A1 A1 (3)
(b)	$x=0.06$ $1-0.12-0.0072-0.000864$ 0.8719	B1 M1 A1 (3)
(c)	$\sqrt{\frac{76}{100}}=\frac{1}{5}\sqrt{19}$ $\sqrt{19}=0.8719\times 5$ 4.360	M1 A1 (2)
		[8]

Part	Mark	Additional Guidance
(a)	M1	For an attempt at a Binomial expansion. A attempt is defined as the following • The expansion must start with 1 • The powers of x must be correct • $-4x$ must be used at least once • The denominators (2! And 3!) must be seen. Accept 2 and 6 $(1-4x)^{\frac{1}{2}}=1+\left(\frac{1}{2}\right)(-4x)+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!}(-4x)^2+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(-4x)^3$
	A1	For at least one term in x correct and fully simplified. $1-2x-2x^2-4x^3$
	A1	For the expansion fully correct and simplified
(b)	B1	For finding the value of $x=0.06$
	M1	For substituting **their** value of x into the expansion **provided** $\lvert x\rvert \leqslant 0.25$ Use of **their expansion** or the correct expansion must be seen explicitly here
	A1	0.8719
(c)	M1	For using their value from (b) in $\sqrt{0.76}=\frac{\sqrt{19}}{5}\Rightarrow\sqrt{19}=5\sqrt{0.76}=5\times\text{'0.8719'}$
	A1	For 4.360 rounded correctly
Penalise rounding once only in this question. Answers must round to the given answers.		

Type MS · Expanding a bracket with a negative or fractional power

Answer 1 of 18 · Question 9 part a · 3 marks

2020 JAN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 2 of 18 · Question 9 part a · 3 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

9 (a)	$(1-2x)^{-\frac{1}{2}} = 1+\left(-\frac{1}{2}\times -2x\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(-2x)^2}{2!}+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)(-2x)^3}{3!}+\ldots$ $(1-2x)^{-\frac{1}{2}} = 1+x+\frac{3}{2}x^2+\frac{5}{2}x^3+\ldots$.	M1 A1A1 [3]
(b)	$\frac{1}{\sqrt{0.96}} = \frac{1}{\sqrt{\frac{96}{100}}} = \frac{10}{4\sqrt{6}}\times\frac{\sqrt{6}}{\sqrt{6}} = \ldots$ $\frac{10\sqrt{6}}{4\times 6} = \frac{5\sqrt{6}}{12}$ *	M1 A1 cso [2]
ALT – confirming given result		
	$\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12} \Rightarrow 12 = \sqrt{0.96}\times 5\sqrt{6}$ $12^2 = 0.96\left(5\sqrt{6}\right)^2 = 0.96\times 5^2\times 6$ *	[M1 A1cso]
(c)	$\frac{1}{\left(5\sqrt{6}-12\right)}\times\frac{\left(5\sqrt{6}+12\right)}{\left(5\sqrt{6}+12\right)}$ $= \frac{5\sqrt{6}+12}{150-12^2} = \frac{5\sqrt{6}+12}{6} = \frac{5\sqrt{6}}{6}+2$	M1 A1 [2]
(d)	$1-2x = 0.96 \Rightarrow 2x = 0.04 \Rightarrow x = 0.02$ $\frac{9}{5\sqrt{6}-12} = 9\left(2\times\left[\frac{5\sqrt{6}}{12}\right]+2\right) =: 9\times\left[2\left(1+0.02+\frac{3}{2}\times 0.02^2+\frac{5}{2}\times 0.02^3\right)+2\right] = \ldots$ 36.37116	B1 M1:M1 A1 [4]
		Total 11 marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 2 of 18 · Question 9 part a · 3 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

		$(1-2x)^{-\frac{1}{2}} = 1 + x + \frac{3}{2}x^2 + \frac{5}{2}x^3 + \ldots$
	A1	Fully correct simplified expansion as shown above.
(b)	**M1**	For changing 0.96 to $\frac{96}{100}$ or equivalent fraction and attempting to multiply numerator and denominator by either $\sqrt{6}$ $\left(\text{or } \sqrt{96}\right)$ $\frac{1}{\sqrt{0.96}} = \frac{1}{\sqrt{\frac{96}{100}}} = \frac{10}{4\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \ldots$
	A1 cso	For the correct answer as shown with no errors $\frac{10\sqrt{6}}{4\times 6} = \frac{5\sqrt{6}}{12}$*
ALT – confirming given result		
	M1	For rearranging and squaring OR for squaring on both sides
	A1 cso	For showing that the two sides of the result are equal $12^2 = 0.96\left(5\sqrt{6}\right)^2 = 0.96 \times 5^2 \times 6 = 144$
(c)	**M1**	For multiplying numerator and denominator by $5\sqrt{6}+12$ $\frac{1}{\left(5\sqrt{6}-12\right)} \times \frac{\left(5\sqrt{6}+12\right)}{\left(5\sqrt{6}+12\right)}$ [Can be implied by $\frac{\left(5\sqrt{6}+12\right)}{\left(150-12^2\right)}$ seen]
	A1	For a **correct** expansion of brackets throughout. $\frac{1}{(5\sqrt{6}-12)} \times \frac{(5\sqrt{6}+12)}{(5\sqrt{6}+12)}\left[= \frac{5\sqrt{6}+12}{150-12^2} =\right] \frac{5\sqrt{6}+12}{6} = \frac{5\sqrt{6}}{6} + 2$
(d)	**B1**	For finding the required value of x $1-2x = 0.96 \Rightarrow 2x = 0.04 \Rightarrow x = 0.02$
	M1	For substituting their value of x **provided** it is $-\frac{1}{2} < x < \frac{1}{2}$ into the expansion as follows: $\frac{9}{5\sqrt{6}-12} = 9\left(2 \times \left[\frac{5\sqrt{6}}{12}\right] + 2\right) =$ **Note:** Must show substitution if x is incorrect.
	M1	For substituting their expansion for $\frac{5\sqrt{6}}{12}$ $9 \times \left[2\left(1 + 0.02 + \frac{3}{2} \times 0.02^2 + \frac{5}{2} \times 0.02^3\right) + 2\right] = \cdots$
	A1	For the value of 36.37116 [The calculator value is 36.37117]

Type MS · Expanding a bracket with a negative or fractional power

Answer 2 of 18 · Question 9 part a · 3 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 3 of 18 · Question 6 part b · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

6 a	$p=2 \quad q=\frac{3}{8}$	B1 B1 (2)
b	$(2)\left[1+\frac{1}{3}\left("\frac{3}{8}"x\right)+\frac{\frac{1}{3}\left(-\frac{2}{3}\right)}{2!}\left("\frac{3}{8}"x\right)^2\right]=2+\frac{1}{4}x-\frac{1}{32}x^2$	M1 A1ft A1 (3)
c	$\left(\sqrt[3]{9}=\sqrt[3]{8+1}\Rightarrow x=\frac{1}{3}\right)$ So $"2"+"\frac{1}{4}"\times"\frac{1}{3}"-"\frac{1}{32}"\times\left("\frac{1}{3}"\right)^2\left(=\frac{576+24-1}{288}\right)=\frac{599}{288}$ *	M1 A1 cso (2)
		Total 7 marks

Part	Mark	Additional Guidance
(a)	**B1**	For $p=2$ **or** $q=\frac{3}{8}$ condone $8^{\frac{1}{3}}$
	B1	For $p=2$ **and** $q=\frac{3}{8}$ condone $8^{\frac{1}{3}}$
(b)	**M1**	For an attempt to expand $(1+qx)^{\frac{1}{3}}$ with their value of q up to the term in x^2 It is not necessary to see p at this stage. The definition of an attempt is as follows: • The first term must be 1 • The next term must be correct for their value of q • The powers of qx must be correct eg $(qx)^2$ • The denominators must be correct Simplification not required. Do not allow missing brackets unless recovered later – this is a general point of marking. Ignore any terms with powers higher than 2.
	A1ft	For both algebraic terms fully correct and unsimplified in the expansion of $(1+qx)^{\frac{1}{3}}$ for their value of q. It is not necessary to see p at this point. Ignore any terms with powers higher than 2.
	A1	For all 3 terms correct, all simplified. Ignore any terms with powers higher than 2
If there are any other methods used – send this to review please.		
(c)	**M1**	For correct substitution of $x=\frac{1}{3}$ into their expansion, which must have at least 2 terms, including a term in x^2
	A1cso*	Obtains the given result

Type MS · Expanding a bracket with a negative or fractional power

Answer 3 of 18 · Question 6 part b · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks
	-6 -5 -4 –3 –2 –1 0	B2

Type MS · Expanding a bracket with a negative or fractional power

Answer 4 of 18 · Question 6 part b · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

6 (a)	$\sqrt{4-x}=2\left(1-\frac{x}{4}\right)^{\frac{1}{2}}$ or $A=2, B=4$	B1B1 [2]
(b)	$2\left\{1+\left(\frac{1}{2}\right)\left(-\frac{x}{4}\right)+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{x}{4}\right)^2}{2!}+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{x}{4}\right)^3}{3!}+...\right\}$ $=2-\frac{x}{4}-\frac{x^2}{64}-\frac{x^3}{512}+...$	M1A1ft A1 [3]
(c)	$\frac{\sqrt{305}}{10}=\sqrt{4-x}\Rightarrow x=0.95$ $2-\frac{0.95}{4}-\frac{0.95^2}{64}-\frac{0.95^3}{512}=1.7467238.....\approx 1.74672$	B1 M1A1 [3]
		Total 8 marks

Part	Mark	Additional Guidance
(a)	B1B1	B1B0 for one correct value, B1B1 For the correct values of both A and B A and B must be integers from correct working **Do not accept** $A=4^{\frac{1}{2}}$ or $\sqrt{4}$
(b)	M1	Attempt to expand their $'2'\left(1-\frac{x}{'4'}\right)^{\frac{1}{2}}$ up to x^3 term (with or without '2') Must have: • First term is 1 for their $\left(1-\frac{x}{'4'}\right)^{\frac{1}{2}}$ expansion • Correct powers of $\pm\frac{x}{'4'}$ for each term No simplification needed. Ignore terms beyond x^3
	A1ft	Two correct algebraic terms in their expansion, with or without'2' Not necessarily in lowest terms. Ignore terms beyond x^3.
	A1	All terms correct and in lowest terms. Ignore terms beyond x^3.
(c)	B1	Identifies $x=0.95$ oe. e.g. $\frac{19}{20}$
	M1	Substitutes their x which must come from some working (provided $\lvert x\rvert<4$) into each of the algebraic terms of their expansion.
	A1	1.74672 (must be 5dp) **Note:** Calculator value for $\frac{\sqrt{305}}{10}$ is 1.74642492

Type MS · Expanding a bracket with a negative or fractional power

Answer 4 of 18 · Question 6 part b · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 5 of 18 · Question 6 part b · 3 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

6 (a)	$\sqrt{9-x}=3\left(1-\frac{x}{9}\right)^{\frac{1}{2}} \Rightarrow p=3, \quad q=-\frac{1}{9}$	B1B1 [2]
(b)	$3\left(1-\frac{x}{9}\right)^{\frac{1}{2}}=$ $3\left\{1+\left(\frac{1}{2}\right)\left(-\frac{x}{9}\right)+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{x}{9}\right)^2}{2!}+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{x}{9}\right)^3}{3!}+\ldots\right\}$ $=3\left(1-\frac{x}{18}-\frac{x^2}{648}-\frac{x^3}{11664}\right)=3-\frac{x}{6}-\frac{x^2}{216}-\frac{x^3}{3888}$	M1 A1A1 [3]
(c)	$\frac{31}{4}=9-x \Rightarrow x=\frac{5}{4}$ $\sqrt{\frac{31}{4}} \approx 3-\frac{1.25}{6}-\frac{1.25^2}{216}-\frac{1.25^3}{3888}=2.783930523\ldots \approx 2.78393$	B1 M1A1 [3]
		Total 8 marks

(a)	B1	$p=3$ or $\sqrt{9}$ or $9^{\frac{1}{2}}$	May be shown in their $3\left(1-\frac{x}{9}\right)^{\frac{1}{2}}$ (allow isw)
	B1	$q=-\frac{1}{9}$	
(b)	M1	Attempt to use the binomial expansion for their $(1+qx)^{\frac{1}{2}}$. Must have first term 1, three more terms with ascending powers of x, 2 or 2! and 6 or 3! Seen, and their $\left(-\frac{x}{9}\right)$ used at least once. No simplification needed. Ignore terms beyond x^3	
	A1	Two algebraic terms correct in the expansion of $3\left(1-\frac{x}{9}\right)^{\frac{1}{2}}$. Must be single fractions, not necessarily in lowest terms. Ignore terms beyond x^3	
	A1	All four terms correct and in lowest terms. Ignore terms beyond x^3	
(c)	B1	Identify $x=\frac{5}{4}$ as the value needed to make $\sqrt{9-x}=\sqrt{\frac{31}{4}}$	
	M1	Substitute their x into each of the algebraic terms of their expansion of $\sqrt{9-x}$	
	A1	2.78393 **given to 5DP**.	

Type MS · Expanding a bracket with a negative or fractional power

Answer 5 of 18 · Question 6 part b · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 6 of 18 · Question 1 part a · 3 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

1(a)	$\left(1+\frac{x}{4}\right)^8 = 1+2x+\frac{7}{4}x^2+\frac{7}{8}x^3$ oe	B1M1A1 [3]
(b)	$1+\frac{x}{4}=1.035 \Rightarrow x=0.14$ oe $(1.035)^8 = 1+2(0.14)+1.75(0.14)^2+0.875(0.14)^3 = 1.3167$ [NB: the calculator value is 1.316809037]	B1 M1A1 [3]
		Total 6 marks

Question	Notes	Marks
1	$\left(1+\frac{x}{4}\right)^8$	
(a)	For an attempt at the binomial expansion of $\left(1+\frac{x}{4}\right)^8$ $\left(1+\frac{x}{4}\right)^8 = 1+2x+...$	B1
	For either correct third or fourth term. Need not be simplified. $...+C_2^8\times\left(\frac{x}{4}\right)^2$ or $C_3^8\times\left(\frac{x}{4}\right)^3$	M1
	For the correct expansion fully simplified. $\left(1+\frac{x}{4}\right)^8 = 1+2x+\frac{7}{4}x^2+\frac{7}{8}x^3$ OR $\left(1+\frac{x}{4}\right)^8 = 1+2x+1.75x^2+0.875x^3$ Do not ignore subsequent incorrect simplification.	A1 [3]
(b)	Finds the value of x to be substituted. $1+\frac{x}{4}=1.035 \Rightarrow x=0.14$ oe	B1
	Substitutes their value of x into their expansion provided it has at least 2 terms in x. *Their* $x \neq 1.035$ $(1.035)^8 = 1+2(0.14)+1.75(0.14)^2+0.875(0.14)^3$	M1
	For the correct value of $(1.035)^8 = 1.3167$ rounded correctly. [NB: the calculator value is 1.316809037]	A1 [3]
		Total 6 marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 6 of 18 · Question 1 part a · 3 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 7 of 18 · Question 3 part a · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

3(a)	$\dots \qquad 2! \qquad 3!$ $+\frac{(-5)(-6)(-7)(-8)(px)^4}{4!}+\dots$ $=1-5px+15p^2x^2-35p^3x^3+70p^4x^4+\dots$	M1 A1A1 (3)
(b)	$70p^4+2\times35p^3=0$ $p=-1$	M1 A1 (2) [5]

(a)	
M1	Attempt the binomial expansion up to and including the term in x^4. Must start with 1 and (px) must appear in at least one term. Ignore terms beyond x^4. 2! or 2, 3! or 6, 4! or 24 accepted.
A1	Any 2 correct algebraic terms, simplified (1 is not algebraic) Numbers must be simplified but $(px)^n$, $n = 2,3,4$ allowed
A1	Fully correct simplified expansion as shown but allow terms such as +(- 5px) etc
(b)	
M1	Use their coefficients and the **given** equation to form an equation in p (If powers of x included give M0)
A1	Correct value of p $p=-1$ only Must have come from correct working

Type MS · Expanding a bracket with a negative or fractional power

Answer 7 of 18 · Question 3 part a · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	16	

Type MS · Expanding a bracket with a negative or fractional power

Answer 8 of 18 · Question 9 part a · 3 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

9(a)	$(1+2x)^{-\frac{1}{3}} = 1+\left(-\frac{1}{3}\right)(2x)+\frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)(2x)^2}{2!}+\frac{\left(-\frac{1}{3}\right)\left(-\frac{4}{3}\right)\left(-\frac{7}{3}\right)(2x)^3}{3!}+\ldots$ $=1-\frac{2x}{3}+\frac{8x^2}{9}-\frac{112x^3}{81}$	M1 A1A1 [3]
(b)	$-\frac{1}{2} < x \leqslant \frac{1}{2}$ accept $-\frac{1}{2} < x < \frac{1}{2}$	B1 [1]
(c)	$f(x) = (2+kx^2)\left(1-\frac{2x}{3}+\frac{8x^2}{9}-\frac{112x^3}{81}\right)$ $=2-\frac{4x}{3}+\frac{16}{9}x^2+kx^2-\frac{224}{81}x^3-\frac{2k}{3}x^3$	M1 M1A1 [3]
(d)	$-\frac{224}{81}-\frac{2k}{3}=-\frac{8}{3} \Rightarrow k=-\frac{4}{27}$	M1A1 [2]
(e)	$\int_{0.1}^{0.2}\left(2-\frac{4x}{3}+\frac{44x^2}{27}-\frac{8}{3}x^3\right)dx=\left[2x-\frac{4x^2}{2\times3}+\frac{44x^3}{3\times27}-\frac{8x^4}{4\times3}\right]_{0.1}^{0.2}$ $=\left(2\times0.2-\frac{2\times0.2^2}{3}+\frac{44\times0.2^3}{81}-\frac{2}{3}\times0.2^4\right)-\left(2\times0.1-\frac{2\times0.1^2}{3}+\frac{44\times0.1^3}{81}-\frac{2}{3}\times0.1^4\right)$ $=0.1828$ NB: Calculator value is 0.18301744	B1FT M1A1ft M1 A1 [5]
		Total 14 marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 8 of 18 · Question 9 part a · 3 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For an attempt at binomial expansion. • The first term is 1 • The denominators are correct. • The powers of x are correct.
	A1	For at least one algebraic term correct and simplified.
	A1	For a fully correct and simplified expansion.
(b)	B1	For the correct range. Accept $\lvert x\rvert < \frac{1}{2}$ Do not accept $-\frac{1}{2} \leq x \leq \frac{1}{2}$
(c)	M1	For showing the intent to multiply their expansion by $\left(2+kx^2\right)$
	M1	For expanding the two brackets up to terms in x^3. Ignore terms in higher powers.
	A1	For the correct expansion. This does not have to be simplified. Must be in ascending powers, but accept $\frac{16}{9}x^2$ and kx^2 in either order and accept $-\frac{224}{81}x^3$ and $-\frac{2k}{3}x^3$ in either order. $f(x) = 2 - \frac{4x}{3} + x^2\left(\frac{16}{9}+k\right) + x^3\left(-\frac{224}{81}-\frac{2k}{3}\right)$ ISW errors in collection of coefficients once correct expansion in ascending powers seen.
(d)	M1	For setting their coefficient of x^3 equal to $-\frac{8}{3}$
	A1	For the correct value of k
(e)	B1FT	For substituting in their k into coefficient for x^2 in their expansion from (c) and having the coefficient of x^3 as $-\frac{8}{3}$. If the coefficient of x^2 is not correct then the substitution must be seen to award this mark. Note: this is an M mark in epen.
	M1	For an attempt to integrate their expansion which must have 4 terms. At least two powers of x to increase by 1 (including $c \rightarrow cx$). No power of x to reduce.
	A1ft	For a correctly integrated expression follow through their 4 term polynomial. Simplified integrated expression is $2x - \frac{2x^2}{3} + \frac{44x^3}{81} - \frac{2}{3}x^4$
	M1	For substituting in 0.2 and 0.1 the correct way around in their integrated expression. Expression should not have been differentiated and needs to have changed. Not dependent on first M. Must show the substitution if their integral or their limits are incorrect. Correct awrt 0.1828 following correct integral with correct limits implies this mark.
	A1	For the correct value of awrt 0.1828 NB: The calculator value is 0.18301744

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2023 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 9 of 18 · Question 2 part a · 4 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

2(a)	$\frac{2}{\sqrt{1+3x}} = 2(1+3x)^{\left(-\frac{1}{2}\right)}$	B1
	$\{2\}\left(1+\left(-\frac{1}{2}\right)(3x)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(3x)^2}{2!}+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)(3x)^3}{3!}+\ldots\right)$	M1A1
	$2-3x+\frac{27x^2}{4}-\frac{135x^3}{8}+\ldots$	A1 [4]
(b)	$-\frac{1}{3}<x<\frac{1}{3}$ (Accept $\lvert x\rvert<\frac{1}{3}$)	B1 [1]
		Total 5 marks

Part	Mark	Notes
(a)	**B1**	Correct simplification of the given expression. This mark may be implied by a correct expansion.
	M1	Attempts at the binomial expansion $(1+3x)^{\pm\frac{1}{2}}$ or $2(1+3x)^{\pm\frac{1}{2}}$ For an attempt at the binomial expansion. • The first term is 1 or 2 • The powers of $3x$ are correct in all terms, e.g. $(3x)^2$ • The correct denominators are used, 2! and 3! oe
	A1	Allow this mark for at least 1 correct algebraic term, correctly simplified, from $-3x+\frac{27x^2}{4}-\frac{135x^3}{8}$ or $-\frac{3}{2}x+\frac{27x^2}{8}-\frac{135x^3}{16}$
	A1	Correct simplified expansion in ascending order, coefficients must be in simplest **fractions**. (Ignore extra terms with powers > 3) Do not isw
(b)	**B1**	For the correct range of values of x. $-\frac{1}{3}<x<\frac{1}{3}$ or $\lvert x\rvert<\frac{1}{3}$, do not accept $\lvert 3x\rvert<1$

ALT for (a) Uses **Maclaurin's expansion (If seen send to review)**

$$f(x) = f(0) + f'(0)x + \frac{f''(x)}{2!}x^2 + \frac{f'''(x)}{3!}x^3 + \ldots$$

B1 Correct simplification of the given expression.

M1 Achieves $f'(x) = P(1+3x)^{-\frac{3}{2}}, f''(x) = Q(1+3x)^{-\frac{5}{2}}, f'''(x) = R(1+3x)^{-\frac{7}{2}}, P, Q, R \neq 0$ and attempts to find the values of $f'(0)$, $f''(0)$ and $f'''(0)$

A1 Correct unsimplified expansion OR at least 2 correct simplified terms

A1 Fully correct simplified expansion. (Ignore extra terms with powers higher than 3)

For reference, the correct derivatives are:

$$f'(x) = -3(1+3x)^{-\frac{3}{2}}, f''(x) = \frac{27}{2}(1+3x)^{-\frac{5}{2}}, f'''(x) = \frac{-405}{4}(1+3x)^{-\frac{7}{2}}$$

Type MS · Expanding a bracket with a negative or fractional power

Answer 9 of 18 · Question 2 part a · 4 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and c, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

Method marks for differentiation and integration

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Use of a formula

Generally, the method mark is gained by **either** quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Type MS · Expanding a bracket with a negative or fractional power

Answer 10 of 18 · Question 1 part whole · 3 marks

2025 NOV · Paper 01 · extract 2 of 3 · find part whole in this mark scheme extract

Exact answers
When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)
Penalise only once per question for failing to round as instructed – i.e. giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the is rule may allow the mark to be awarded before the final answer is given.

Type MS · Expanding a bracket with a negative or fractional power

Answer 10 of 18 · Question 1 part whole · 3 marks

2025 NOV · Paper 01 · extract 3 of 3 · find part whole in this mark scheme extract

Question	Scheme	Marks
1	$\left(1-\frac{x}{4}\right)^{-2} = 1+(-2)\left(-\frac{x}{4}\right)+\frac{(-2)(-3)\left(-\frac{x}{4}\right)^2}{2!}+\frac{(-2)(-3)(-4)\left(-\frac{x}{4}\right)^3}{3!}$	M1
	$= 1+\frac{x}{2}+\frac{3x^2}{16}+\frac{x^3}{16}$	A1A1 [3]
		Total 3 marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 11 of 18 · Question 5 part b · 3 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

5 a	$(2+3x)^{-1}=\frac{1}{2}\left(1+\frac{3}{2}x\right)^{-1}$ so $p=\frac{1}{2}$ and $q=\frac{3}{2}$ oe	B1 B1 (2)
b	$\left(\frac{1}{2}\right)\left[1+(-1)\left("\frac{3}{2}"x\right)+\frac{(-1)(-2)}{2!}\left("\frac{3}{2}"x\right)^2+\frac{(-1)(-2)(-3)}{3!}\left("\frac{3}{2}"x\right)^3+\ldots\right]$	M1 A1ft
	$\frac{1}{2}-\frac{3}{4}x+\frac{9}{8}x^2-\frac{27}{16}x^3+\ldots$	A1 (3)
c	$\frac{1}{2}-\frac{3}{4}x+\frac{9}{8}x^2-\frac{27}{16}x^3+\frac{1}{2}x-\frac{3}{4}x^2+\frac{9}{8}x^3+\ldots$	M1
	$\frac{1}{2}-\frac{1}{4}x+\frac{3}{8}x^2-\frac{9}{16}x^3$	A1 (2)
d	$\int_0^{0.5}\left(\frac{1}{2}-\frac{1}{4}x+\frac{3}{8}x^2-\frac{9}{16}x^3\right)dx=\frac{1}{2}x-\frac{1}{8}x^2+\frac{1}{8}x^3-\frac{9}{64}x^4$	M1 A1
	$\frac{1}{2}\left(\frac{1}{2}\right)-\frac{1}{8}\left(\frac{1}{2}\right)^2+\frac{1}{8}\left(\frac{1}{2}\right)^3-\frac{9}{64}\left(\frac{1}{2}\right)^4=0.2256$ for awrt 0.2256	M1 A1 (4)
		Total 11 marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 11 of 18 · Question 5 part b · 3 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Marks	Notes
(a)	B1	For $p=\frac{1}{2}$ or 2^{-1} or $q=\frac{3}{2}$
	B1	For $p=\frac{1}{2}$ or 2^{-1} and $q=\frac{3}{2}$
		NB $\frac{1}{2}\left(1+\frac{3}{2}x\right)^{-1}$ scores B1 B1
(b)	M1	For an attempt to expand $(1+qx)^{-1}$ with their value of q up to the term in x^3 It is not necessary to see p at this stage. The definition of an attempt is as follows: • The first term must be 1 • The next term must be correct for their value of q • The powers of qx must be correct eg $(qx)^2$ • The denominators must be correct Simplification not required. Do not allow missing brackets unless recovered later – this is a general point of marking.
	A1ft	For at least 3 terms fully correct and unsimplified for their value of q. It is not necessary to see p at this point.
	A1	For all 4 terms correct, all simplified.
	If there are any other methods used – send this to review please.	
(c)	M1	For multiplying their expansion by $(1+x)$ There must be a clear attempt to multiply to get 7 terms, allow up to 2 errors. Ignore terms with powers higher than 3.
	A1	For all 4 terms correct, ignore terms with powers higher than 3.
(d)	M1	For a minimally acceptable attempt to integrate an expression with at least 3 terms, which must include a term in x^3 See general guidance for definition of minimally acceptable attempt.
	A1ft	For a fully correct integration of their expression (defined as above for the M mark).
	M1	For substitution of limits into a changed expression, the correct way round – this can be implied if the final answer is correct. We don't need to see 0 substituted in.
	A1	For awrt 0.2256 (Note the calculator value is 0.2288461……)

Type MS · Expanding a bracket with a negative or fractional power

Answer 11 of 18 · Question 5 part b · 3 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
	0 *0.25* *0.5* **1** *1.5* **2** 3	B2

Type MS · Expanding a bracket with a negative or fractional power

Answer 12 of 18 · Question 7 part a · 3 marks

2024 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

7(a)	$\left(1+2x^2\right)^{-\frac{3}{4}} = 1+\left(-\frac{3}{4}\right)\left(2x^2\right)+\frac{\left(-\frac{3}{4}\right)\left(-\frac{7}{4}\right)\left(2x^2\right)^2}{2!}+\frac{\left(-\frac{3}{4}\right)\left(-\frac{7}{4}\right)\left(-\frac{11}{4}\right)\left(2x^2\right)^3}{3!}$ $= 1-\frac{3}{2}x^2+\frac{21}{8}x^4-\frac{77}{16}x^6$	M1 A1A1 [3]
(b)	$f(x) = (2+kx)\left(1-\frac{3}{2}x^2+\frac{21}{8}x^4-\frac{77}{16}x^6\right)$ $= 2+kx-3x^2-\frac{3k}{2}x^3+\frac{21}{4}x^4+\frac{21k}{8}x^5$	 M1A1 [2]
(c)	$14\times(-3) = \frac{21k}{8} \Rightarrow k = \frac{14\times(-3)\times 8}{21} = -16$	M1A1 [2]
		Total 7 marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 12 of 18 · Question 7 part a · 3 marks

2024 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Type MS · Expanding a bracket with a negative or fractional power

Answer 13 of 18 · Question 9 part b · 3 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

9 (a)	$\frac{1}{(2-x)^3} = (2-x)^{-3} = \frac{1}{8}\left(1-\frac{x}{2}\right)^{-3} \Rightarrow p = \frac{1}{8},\ q = \frac{1}{2}$	B1B1 [2]
(b)	$\frac{1}{8}\left(1-\frac{x}{2}\right)^{-3} = \frac{1}{8}\left[1+(-3)\left(-\frac{x}{2}\right)+\frac{(-3)(-4)\left(-\frac{x}{2}\right)^2}{2!}+\frac{(-3)(-4)(-5)\left(-\frac{x}{2}\right)^3}{3!}\right]$ $= \frac{1}{8}+\frac{3}{16}x+\frac{3}{16}x^2+\frac{5}{32}x^3+\ldots$	M1 A1A1 [3]
(c)	$(a+bx)\left(\frac{1}{8}+\frac{3}{16}x+\frac{3}{16}x^2\right) = \frac{a}{8}+x\left(\frac{2b}{16}+\frac{3a}{16}\right)+\left\{x^2\left(\frac{3a}{16}+\frac{3b}{16}\right)\right\}$ $\Rightarrow \frac{3}{8} = \frac{a}{8} \Rightarrow a = 3$ $\Rightarrow -\frac{43}{16} = \frac{2b+3a}{16} \Rightarrow 2b = -43-9 = -52 \Rightarrow b = -26$	M1 A1 A1 [3]
(d)	$\frac{3a+3b}{16} = \frac{9-78}{16} = -\frac{69}{16}$ oe	M1A1 [2]
		Total 10 marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 13 of 18 · Question 9 part b · 3 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

(b)	
M1	Attempts to use the binomial expansion for their $(1-qx)^{-3}$. Must have first term 1, three more terms with ascending powers of x, 2 or 2! and 6 or 3! seen , and their $\left(-\frac{x}{2}\right)$ used at least once. No simplification needed. Ignore terms beyond x^3
A1	Two algebraic terms correct in the expansion for their $(1-qx)^{-3}$.. Must be single fractions, not necessarily in lowest terms. Ignore terms beyond x^3
A1	All four terms correct and in lowest terms. Ignore terms beyond x^3
(c)	
M1	For either their $\frac{1}{8}a=\frac{3}{8}$ or their $\frac{3}{16}a+$ their $\frac{1}{8}b=-\frac{43}{16}$ May be implied by a correct value of a or b.
A1	$a=3$
A1	$b=-26$ **NB** answers $a=3$ and $b=-26$ scores 3/3
(d)	
M1	Substituting their a and their b into their $\frac{3a+3b}{16}$
A1	$-\frac{69}{16}$ oe **NB** If $p=2$, $q=\frac{1}{2}$ is used then $a=\frac{3}{16}$ and $b=-\frac{13}{8}$ which substituted into $3a+3b$ gives an answer of $-\frac{69}{16}$ but scores A0

Type MS · Expanding a bracket with a negative or fractional power

Answer 13 of 18 · Question 9 part b · 3 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

(b)	M1	For attempt to integrate one of: $9-9e^{-3x}$ or $e^{3x}-1$ or $\pm[(9-9e^{-3x})-(e^{3x}-1)]$ Limits may not be present. At least one term correct. Ignore +c if included.
	$A1_2$	For correct integration of one of the exponential terms $\pm 9e^{-3x} \rightarrow \mp\frac{9}{3}e^{-3x}$ **or** $\pm e^{3x} \rightarrow \pm\frac{1}{3}e^{3x}$ Limits need not be present. Ignore +c if included.
	$A1_3$	For correct integration of both curves $9x+3e^{-3x}$ **and** $\frac{e^{3x}}{3}-x$ or for a **fully correct** integration where the difference between two expressions is found $\pm\left(10x-\frac{9}{-3}e^{-3x}-\frac{e^{3x}}{3}\right)$ or $\pm\left(9x-\frac{9}{-3}e^{-3x}-\frac{e^{3x}}{3}+x\right)$ **Note: this is an M mark in epen**
	$M1_4$	For the difference between the two expressions either before or after integration. Allow subtraction either way around. **Note: this is an A mark in epen**
	$dM1_5$	Substitution of correct limits into their integrated expressions (limits subtracted the correct way around). Dependent on first M scored. If substituting before difference found then must substitute into both integrated expressions. May be implied by awrt 1.99 If integration is not correct then substitution must be shown.
	$A1_6$	For the correct answer oe. Must be exact value.

Type MS · Expanding a bracket with a negative or fractional power

Answer 14 of 18 · Question 9 part b · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
9(a)	$(3(3-x)^{-3} = \quad \frac{1}{9}\left(1-\frac{x}{3}\right)^{-3}\ )$ $\quad a=\frac{1}{9} \quad b=\frac{1}{3}$	B1 B1 [2]
(b)	$\left(1-\frac{x}{3}\right)^{-3} =$ $\left[1+(-3)\left(-\frac{x}{3}\right)+\frac{(-3)(-4)\left(-\frac{x}{3}\right)^2}{2!}+\frac{(-3)(-4)(-5)\left(-\frac{x}{3}\right)^3}{3!}\right]$	M1 A1ft
	$\frac{1}{9}+\frac{1}{9}x+\frac{2}{27}x^2+\frac{10}{243}x^3$	A1 [3]
(c) (i)	$\frac{24}{125}=\frac{3}{(3-x)^3}$ or $\frac{125}{8}=(3-x)^3 \Rightarrow \frac{5}{2}=3-x$	
	$x=0.5$ oe	B1
	$\frac{1}{9}+\frac{1}{9}("0.5")+\frac{2}{27}("0.5")^2+\frac{10}{243}("0.5")^3 \quad (=0.19033)$	B1ft
(ii)	$\pm\left(\frac{\frac{24}{125}-"0.19033"}{\frac{24}{125}}[\times 100]\right)$ oe	M1
	0.87% or −0.87%	A1 [4]
		Total 9 marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 14 of 18 · Question 9 part b · 3 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 15 of 18 · Question 10 part b · 3 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

10 (a)	$\frac{3}{\sqrt{9-3x}}=\frac{1}{\sqrt{9}}\times\frac{3}{\sqrt{1-\frac{3x}{9}}}=\frac{1}{\sqrt{1-\frac{x}{3}}}=\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$ *	B1B1* cso [2]
(b)	$\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}=1+\left(-\frac{1}{2}\right)\left(-\frac{x}{3}\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{x}{3}\right)^2}{2!}+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)\left(-\frac{x}{3}\right)^3}{3!}$ $\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}=1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}$	M1 A1A1 [3]
(c)	$3\mathrm{f}(x)=\frac{3(1+2x)}{\sqrt{9-3x}}=\frac{3(1+2x)}{3\sqrt{1-\frac{x}{3}}}=(1+2x)\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$ $3\mathrm{f}(x)=(1+2x)\left(1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}\right)$ $3\mathrm{f}(x)=(1+2x)\left(1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}\right)=1+\frac{13}{6}x+\frac{3}{8}x^2+\frac{41}{432}x^3$	B1 M1 M1A1 [4]
(d)	$\int_{0.1}^{0.2}\frac{3+6x}{\sqrt{9-3x}}\,\mathrm{d}x=\int_{0.1}^{0.2}\left(1+\frac{13}{6}x+\frac{3}{8}x^2+\frac{41}{432}x^3\right)\mathrm{d}x$ $=\left[x+\frac{13}{12}x^2+\frac{3}{24}x^3+\frac{41}{1728}x^4\right]_{0.1}^{0.2}$ $\int_{0.1}^{0.2}\frac{1+2x}{\sqrt{9-3x}}\,\mathrm{d}x=\frac{1}{3}\left[\left(0.2+\frac{13}{12}\times0.2^2+\frac{3}{24}\times0.2^3+\frac{41}{1728}\times0.2^4\right)-\left(0.1+\frac{13}{12}\times0.1^2+\frac{3}{24}\times0.1^3+\frac{41}{1728}\times0.1^4\right)\right]=0.044\ 470\ 2$	M1A1ft dM1A1 [4]
		Total 13 marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 15 of 18 · Question 10 part b · 3 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

	B1*	For simplifying to the required form $\frac{3}{\sqrt{9-3x}}=\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$
(b)	**M1**	For an attempt to find the binomial expansion for the given expression. • The expansion must begin with 1 • The denominators must be correct (ie. 2! And 3!) on the third and fourth terms. • The power of x must be correct (ie. $\left(-\frac{x}{3}\right)$ $\left(-\frac{x}{3}\right)^2$ and $\left(-\frac{x}{3}\right)^3$ with the correct corresponding denominators.
	A1	For one algebraic term correct and simplified
	A1	For the whole expansion correct up to x^3 Ignore extra terms beyond these. $\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}=1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}$
(c)	**B1**	For writing down $3f(x)=(1+2x)\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$
	M1	For replacing $\left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$ with their $\left(1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}\right)$ provided their expansion has at least two terms one of which is an algebraic term.
	M1	For an attempt to expand $(1+2x)\left('1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}'\right)$ up to and including the term in x^3. Accept the expansion of $(3+6x)\left('1+\frac{x}{6}+\frac{x^2}{24}+\frac{5x^3}{432}'\right)$
	A1	For the correct expansion $3f(x)=1+\frac{13}{6}x+\frac{3}{8}x^2+\frac{41}{432}x^3$ ignore extra terms
(d)	**M1**	For an attempt to integrate their expansion of 3f(x) or f(x) which must have as a minimum two terms, one of which is a term in x, x^2 or x^3 Ignore limits for this mark. They may multiply by $\frac{1}{3}$ at this stage so accept e.g $\left[\frac{x}{3}+\frac{13}{36}x^2+\frac{3}{72}x^3+\frac{41}{5184}x^4\right]$
	A1ft	For a correct integrated expression ft their expression.
	dM1	For substituting both values of 0.2 and 0.1 the correct way around seen. This mark can be implied by the value of 0.044 470 2 seen. If their expansion from (c) has not been divided by 3 for the integration, it must be divided by 3 here for the award of this mark.
	A1	For the correct value of 0.044 470 2 **NB**: The calculator value is 0.044 470 77

Type MS · Expanding a bracket with a negative or fractional power

Answer 15 of 18 · Question 10 part b · 3 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 16 of 18 · Question 7 part a · 3 marks

2023 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

7(a)	$\left(1+\frac{x}{3}\right)^{-3}=\left[1+(-3)\left(\frac{x}{3}\right)+\frac{(-3)(-3-1)}{2!}\left(\frac{x}{3}\right)^2+\frac{(-3)(-3-1)(-3-2)}{3!}\left(\frac{x}{3}\right)^3\ldots\right]$ $=1-x+\frac{2}{3}x^2-\frac{10}{27}x^3$	M1A1 A1 [3]
(b)	$-3<x<3$ or $\lvert x\rvert<3$	B1 [1]
(c)	$\left((3+x)^{-3}=3^{-3}\times\left(1+\frac{x}{3}\right)^{-3}=\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}\right)$ $P=\frac{1}{27}$, $Q=\frac{1}{3}$	 B1, B1 [2]
(d)	$\frac{(1+4x)}{(3+x)^3}=(1+4x)\times\frac{1}{27}\times\left(1-x+\frac{2}{3}x^2\right)$ $=\frac{1}{27}\times\left(1-x+\frac{2}{3}x^2+4x-4x^2+\ldots.\right)$ $=\frac{1}{27}\left(1+3x-\frac{10x^2}{3}\right)$ or $\frac{1}{27}+\frac{x}{9}-\frac{10x^2}{81}$	 M1 A1 [2]
(e)	$\int_0^{0.2}\left(\frac{1}{27}+\frac{x}{9}-\frac{10x^2}{81}\right)\mathrm{d}x=\left[\frac{x}{27}+\frac{x^2}{18}-\frac{10x^3}{243}\right]_0^{0.2}$ $=\left(\frac{0.2}{27}+\frac{0.2^2}{18}-\frac{10\times0.2^3}{243}\right)-[0]$ $=0.0093004$	M1 M1 A1 [3]
		Total 11 marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 16 of 18 · Question 7 part a · 3 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For applying a correct binomial expansion in unsimplified form. Minimum required: • The expansion begins with 1 • The next term is correct • The powers of $\frac{x}{3}$ must be correct eg $\left(\frac{x}{3}\right)^2$ • The denominators are correct. Do not allow missing brackets unless recovered later – this is a general point of marking. Ignore any terms with powers higher than 3.
	A1	Following M1 (this is a general point of marking, A marks can only follow M marks), all conditions above met, must see $1 - x$ and at least the term in x^2 or x^3 correct and simplified. Ignore any terms with powers higher than 3.
	A1	A fully correct and simplified expansion. Ignore any terms with powers higher than 3..
(b)	B1	For the correct validity.
(c)	B1	For the correct value of P or Q explicitly written or embedded in $\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}$
	B1	For the correct values of P and Q explicitly written or embedded in $\frac{1}{27}\times\left(1+\frac{x}{3}\right)^{-3}$
(d)	M1	For attempting to multiply their expansion, which must be of the form "P"× (their expansion from part (a), with a minimum of 3 terms, by $(1 + 4x)$. An attempt must include 3 correctly multiplied out terms of their expansion before simplification. The "P" may remain factorised. Ignore any terms which to powers of x higher than 2.
	A1	For the correct expansion in either form shown. Allow equivalent coefficients. Ignore powers of x higher than 2.
(e)	M1	For an attempt to integrate their expression from part (d), provided it has at least one constant term and at least one algebraic term. See general guidance, no power of x must decrease.
	M1	For substituting in the value of 0.2 into a changed expression and subtracting the correct way. Must see the explicit substitution of 0.2 at least once, if the final answer is not correct. Can be implied if final answer correct. Substitution of 0 does not need to be seen.
	A1	For the value of 0.0093004 [The calculator value is 0.0093316] Accept the value of $\frac{113}{12150}$ if seen.

Type MS · Expanding a bracket with a negative or fractional power

Answer 16 of 18 · Question 7 part a · 3 marks

2023 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 17 of 18 · Question 5 part a · 3 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Number		
5 (a)	$1+\frac{1}{2}(-x)+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}(-x)^2+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(-x)^3$	M1
	$1-\frac{1}{2}x-\frac{1}{8}x^2-\frac{1}{16}x^3$	A1 A1 (3)
(b)	$x=0.08$	B1
	$1-\frac{1}{2}"(0.08)"-\frac{1}{8}"(0.08)"^2-\frac{1}{16}"(0.08)"^3$	M1
	0.959168 cao	A1 cao (3)
(c)	$(\sqrt{0.92}=\frac{\sqrt{23}}{5}\Rightarrow\sqrt{23}=)"0.959168"\times 5$	M1
	4.79584	A1 (2) **[8]**

Part	Mark	Additional Guidance
(a)	M1	For an attempt at a Binomial expansion. An attempt is defined as the following • The expansion must start with 1 • The powers of their $-x$ must be correct • $-x$ must be used at least once • The denominators 2! And 3! must be seen. Accept 2 and 6 Can be implied by at least 2 correct terms in an expansion
	A1	For at least one term in x correct and fully simplified.
	A1	For the expansion fully correct and simplified. Ignore terms in higher powers of x.
(b)	B1	For finding the value of $x = 0.08$
	M1	For correctly substituting **their** value of x into the expansion **provided** $\|x\| < 1$ Use of **their expansion** or the correct expansion must be seen explicitly here
	A1	cao
(c)	M1	For use of their value from (b) in $\sqrt{0.92}=\frac{\sqrt{23}}{5}\Rightarrow\sqrt{23}="0.959168"\times 5$
	A1	Cao
		Allow the final M1 A1 if $\sqrt{0.92}\times 5$ is clearly written and 4.79584 is clearly indicated as the answer.

Type MS · Expanding a bracket with a negative or fractional power

Answer 17 of 18 · Question 5 part a · 3 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 18 of 18 · Question 9 part a · 3 marks

2023 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

9(a)	$\left(1-8x^2\right)^{-\frac{1}{2}}=1+\left(-\frac{1}{2}\right)\left(-8x^2\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-8x^2\right)^2}{2!}$ $+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)\left(-8x^2\right)^3}{3!}$ $\left(1-8x^2\right)^{-\frac{1}{2}}=1+4x^2+24x^4+160x^6+\ldots$	M1 A1A1 [3]
(b)	$\left((a+bx)"\left(1+4x^2+24x^4+160x^6\right)"=\right)\left(a+bx+4ax^2\right)+4bx^3+24ax^4$ $4b=20\Rightarrow b=5$ $24a=48\Rightarrow a=2$	M1A1ft A1 A1 [4]
(c)	$\left(\int_0^{0.2} g(x)\,dx=\int_0^{0.2}\left(\frac{2+5x}{\sqrt{1-8x^2}}\right)dx=\right)$ $\int_0^{0.2}\left("2"+"5"x+"8"x^2+"20"x^3+"48"x^4\right)dx$ $\left(\int_0^{0.2} g(x)\,dx=\right)\left["2x+\frac{5x^2}{2}+\frac{8x^3}{3}+\frac{20x^4}{4}+\frac{48x^5}{5}"\right]_0^{0.2}$ $\left(\int_0^{0.2} g(x)\,dx=\right)"2\times0.2+\frac{5\times0.2^2}{2}+\frac{8\times0.2^3}{3}+\frac{20\times0.2^4}{4}+\frac{48\times0.2^5}{5}"$ $=0.532405333\approx0.5324$ Note: The calculator value is 0.5347698	 M1 M1 M1 A1 [4]
		Total 11 marks

Type MS · Expanding a bracket with a negative or fractional power

Answer 18 of 18 · Question 9 part a · 3 marks

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Part	Mark	Notes
(a)	M1	For an attempt at the binomial expansion. The expansion must: • Begin with 1 • The denominators must be correct • The powers of $-8x^2$ must be correct eg $\left(-8x^2\right)^2$ Do not allow missing brackets unless recovered later – this is a general point of marking. Ignore any terms with powers higher than 3. Allow a misread of $\frac{1}{2}$ for $-\frac{1}{2}$ but **not** $8x^2$ for $-8x^2$ - see gen guidance for misreads.
	A1	Following M1 (this is a general point of marking, A marks can only follow M marks). All conditions above met and at least one **algebraic** term correct and simplified.
	A1	Fully correct and simplified
(b)	M1	For attempting to expand $(a+bx)(1+cx^2+dx^4+.....)$ to reach at least the correct terms in x^3 and x^4 for their expansion from part (a) ie $4bx^3$ and $24ax^4$ or cbx^3 and dax^4 Ignore terms with powers higher than 4. It is only necessary to see the correct terms (for their expansion) in x^3 and x^4. They must be using an expansion from (a) of the form $1+cx^2+dx^4+.....$ This mark may be implied from any following working.
	A1ft	For a fully correct expansion in terms of x, a and b,up to the term in x^4 or for $4b = 20$ AND $24a = 48$. **or** if using an incorrect expansion from part (a) of the form $1+cx^2+dx^4+.....$ for a fully correct expansion **or** $cb = 20$ AND $da = 48$. Ignore terms with powers higher than 4.
	A1ft	Either a or b correct ft their expansion from (a) of the form $1+cx^2+dx^4+.....$ **and** $cb = 20$ AND $da = 48$.
	A1ft	Both a and b correct. ft their expansion from (a) of the form $1+cx^2+dx^4+.....$ **and** $cb = 20$ AND $da = 48$.
(c)	M1	For using their expansion from (b) with the correct limits. Although their expression does not need to be fully correct, it must be clear it has arisen from attempting to find a and b in part (b) and using these values. The expression must include the **first** 5 terms in their expansion of the form $p+qx+rx^2+sx^3+tx^4$ from part (b) using their a and b and may include more for this mark. It is not necessary to see the integral sign, if the candidate clearly integrates later. Similarly, if the limits are not present until substitution, this mark can be awarded.
	M1	For an attempt to integrate an expression (see general guidance) – no power of x to decrease. This is not a dependent method mark and the mark is for an attempt to integrate, **but** they must integrate all their terms and there must be a minimum of 2 algebraic terms.
	M1	For substitution of the given limits the correct way around. There must be at least one correct substitution of limits into their **changed** expression with a minimum of 2 algebraic terms (ie this again is not a dependent method mark). 0 need not be substituted if substitution into the changed expression gives an evaluation of 0. This may be implied by a correct answer, but must be shown if the final answer is not correct.
	A1	For awrt 0.5324 ie condone more decimal places.
The question does not demand that students use algebraic integration or show their working clearly and therefore **a correct answer of 0.5324….** may imply the method marks and full marks should be awarded. A value of 0.5347…..will score 0 marks.		

Type MS · Expanding a bracket with a negative or fractional power

Answer 18 of 18 · Question 9 part a · 3 marks

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Question	Scheme	Marks